

Web Programming Course: Computer-Based Test

Notice :

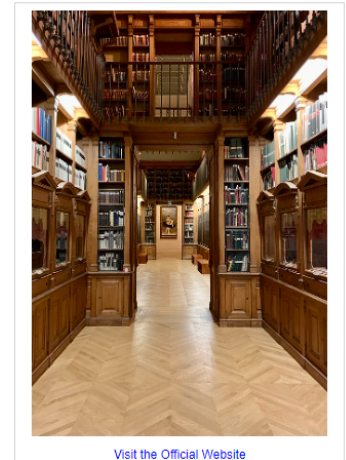
- (1) Please name the html files of your answers to the first six questions as [your student ID-1.html], [your student ID-2.html], and so on. For example, A student with ID 66666666 should create a directory of 66666666 at the location specified by the teaching assistant, and the directory should contain 7 files, including 66666666-1.html, 66666666-2.html, and so on (please write HTML, JavaScript, and CSS in the same file, and do not split it into three files).
- (2) Please do not forget to write the seventh question. Please also save it as [your student ID-7.txt] and put it in the directory of your student ID.
- (3) The total score is 140, and the maximum score you can get is 105 (you can choose preferred questions).

1. (25%) Please develop a webpage using HTML and CSS as follows:

Welcome to the Museum of History

The Museum of History is a place dedicated to preserving and showcasing the rich history of our world. Founded in 1950, our mission is to educate and inspire visitors about the past.

		
Exhibit 1: Ancient Civilizations		
Explore the fascinating artifacts and relics from ancient civilizations, including pottery, tools, and sculptures. Gain insights into the daily life and cultural practices of these remarkable societies.	Discover the architectural marvels and advancements in art from the ancient world. From the Pyramids of Egypt to the Acropolis in Greece, this exhibit showcases the ingenuity of human civilization.	Uncover the mysteries of ancient rituals and beliefs. From religious ceremonies to burial practices, delve into the spiritual aspects that shaped the identity of ancient cultures.
		
Exhibit 2: The Renaissance Era		
Immerse yourself in the vibrant world of the Renaissance. Witness the explosion of creativity in art, science, and literature. Experience the works of Michelangelo, Leonardo da Vinci, and other iconic figures.	Journey through a time of rebirth and transformation. Explore the societal changes, technological innovations, and the birth of the printing press that defined the Renaissance era.	Gain insights into the intellectual revolution that shaped modern thinking. From humanism to the scientific method, this exhibit highlights the Renaissance's profound impact on the course of history.
		
Exhibit 3: Space Exploration		
Embark on a cosmic journey through the history of space exploration. Witness the achievements of pioneers like Yuri Gagarin and Neil Armstrong, who ventured beyond our planet.	Explore the technological advancements that propelled humanity into space. From early rockets to the International Space Station, this exhibit showcases the evolution of space travel.	Experience the awe-inspiring beauty of the cosmos. View stunning images captured by telescopes and spacecraft, and learn about ongoing missions that continue to expand our understanding of the universe.



[Visit the Official Website](https://www.ntm.gov.tw/en/)

HTML Structure:

- A title introducing the museum's name (provided).
- A paragraph describing the history and mission of the museum (provided).
- The left part is a table, using *colspan* to merge appropriate cells (to show a theme image and its title), and showcasing three main exhibits (展覽) of the museum.
- The right part is a div containing at least one image of the museum and a hyperlink linking to the museum's official website (<https://www.ntm.gov.tw/en/>) in a new window.

CSS Styling:

- Set the font "Arial" for the entire page (provided).
- Use the float property to display the table and image side by side.
- Set the width (70%) and background color (PowderBlue) of the table to create the appearance of an exhibition list.
- Set borders (1px solid darkgray) and padding (3px) for the table, table headers, and table cells.
- Set the text-align for table cells to left.
- Set the text color (darkgoldenrod) for exhibit titles.
- Set the width (30%) for images inside the table.
- Set the width (25%) for the right-side <div>.

- Add borders effect (1px solid lightgray) to the right-side <div>.
- Set the text color and hover effect for hyperlinks (from “without underline and in blue” to “with underline and in darkred”).

Responsive Design: please ensure the page displays correctly on small screens (width <= 600px).

- Cancel the effect of “float”.
- Change the width for multiple elements.

Exhibit 3: Space Exploration

Embark on a cosmic journey through the history of space exploration. Witness the achievements of pioneers like Yuri Gagarin and Neil Armstrong, who ventured beyond our planet.	Explore the technological advancements that propelled humanity into space. From early rockets to the International Space Station, this exhibit showcases the evolution of space travel.	Experience the awe-inspiring beauty of the cosmos. View stunning images captured by telescopes and spacecraft, and learn about ongoing missions that continue to expand our understanding of the universe.
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[Visit the Official Website](#)

2. (20%) Please develop a webpage to let users perform the four fundamental arithmetic operations: addition, subtraction, multiplication, and division.

● Please show the following message to let the user choose the operator:

" Please enter an arithmetic operator: 1 represents addition (+), 2 represents subtraction (-), 3 represents multiplication (*), 4 represents division (/):"

這個網頁顯示

Please enter an arithmetic operator: 1 represents addition (+), 2 represents subtraction (-), 3 represents multiplication (*), 4 represents division (/):

確定 取消

If the user input is not a valid option, please show an error message "Invalid input! The option is automatically set to 1!" using a popup window and set the option to 1.

這個網頁顯示

Invalid input! The option is automatically set to 1.

確定

● Let the user input two numbers:

這個網頁顯示

Please enter the first number:

確定 取消

這個網頁顯示

Please enter the second number:

確定 取消

If any input value is not a valid number, please show an error message "Invalid input! The value is automatically set to 1." using a popup window and set the number to 1.

這個網頁顯示

Invalid input! The value is automatically set to 1.

確定

- Finally, the webpage shows the results of the arithmetic operation:

Arithmetic Operation Result:

$$11 * 111 = 1221$$

Another example:

Arithmetic Operation Result:

$$123 / 321 = 0.38$$

Other requirements:

- (1) Please modify the CSS in Test-2-partial.html to:
 - Let the number and operator blocks display **side-by-side**.
- (2) If the operator is division, the result should be rounded to two decimal places (取到小數點後兩位).

Hint:

- Please ensure how to set values to HTML elements using DOM (document object model).
- “Side-by-side” is easy to realize by a CSS property.
- `isFinite()`: returns true if a value is a finite number.

3. (20%) Please use JavaScript, HTML, and CSS to develop a gallery of Doraemon accessories (哆啦A夢道具) based on Test-3-Partial.html.

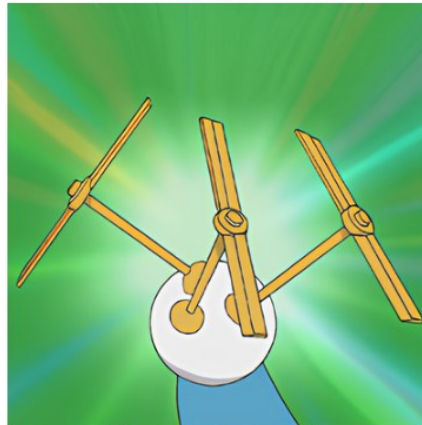
- The initial screen should contain multiple small images below (gadget images) and a large image above (the initial image should be 00.png) °

哆啦A夢道具



- When the user clicks the small image below, the large image above will be replaced with the corresponding gadget image, and the gadget name should be displayed above the large image (點小圖顯示大圖，並顯示名稱).

竹蜻蜓



Other requirements:

- The gadget names **can only be obtained** through the gadgetNames variable.
- The image file names **cannot be modified**.
- The HTML part of “Test-3-partial.html” **cannot be changed**.

4. (20%) Please create a web application that generates random technology-themed articles based on user input. The application should **take a noun as input** and generate an article in two paragraphs in Chinese. Each paragraph should follow a predefined template and incorporate randomly selected phrases, adjectives, objects, verbs, and locations related to technology. Ensure that the articles have a coherent and engaging style. Use string templates to construct the article and display the generated content on the web page.

Your application should include:

- An input field for users to enter a noun.
- A button to trigger the generation of the article.
- A display area to show the two paragraphs of the generated article.

We have the following arrays to hold phrases, adjectives, objects, verbs, and locations.:

```
const phrases = [  
  '不斷地', '科技界認為', '有時候', '不可思議地', '科學家表示',  
  '顯然地', '專家指出', '毫無疑問地', '根據研究', '數據顯示'  
];  
  
const adjectives = [  
  '創新', '前所未有', '具未來性', '革命性', '先進',  
  '突破性', '有效率', '頂尖', '前瞻性', '尖端'  
];  
  
const objects = [  
  '手機', '人工智慧', '雲端技術', '機器人', '虛擬實境',  
  '區塊鏈', '物聯網', '量子電腦', '智慧穿戴', '大數據'  
];  
  
const verbs = [  
  '改變', '優化', '孕育', '塑造', '推動',  
  '革新', '加速', '創造', '引領', '實現'  
];  
  
const locations = [  
  '實驗室裡', '科技公司', '研究中心', '未來世界', '資通訊領域',  
  '創新園區', '科技園區', '智能工廠', '智慧城市', '數位空間'  
];
```


The expected interface: (The font should be set to “王漢宗粗鋼筆體(WCL-07.ttf)”)

廢文產生器

元宇宙

有關元宇宙的消息在科學界掀起了一陣熱潮。有時候，它像前瞻性的人工智慧一樣在智慧城市改變，並且先進地引領著元宇宙產業的未來。許多專家正密切關注元宇宙的最新發展。科技界認為，它不僅是有效率的雲端技術，更是前瞻性地引領著人類的科技進步。

The possible generated content:

- “有關元宇宙的消息在科學界掀起了一陣熱潮。有時候，它像前瞻性的人工智慧一樣在智慧城市改變，並且先進地引領著元宇宙產業的未來。
許多專家正密切關注元宇宙的最新發展。科技界認為，它不僅是有效率的雲端技術，更是前瞻性地引領著人類的科技進步。”
- “有關元宇宙的消息在科學界掀起了一陣熱潮。數據顯示，它像具未來性的大數據一樣在科技公司塑造，並且先進地引領著元宇宙產業的未來。
許多專家正密切關注元宇宙的最新發展。科技界認為，它不僅是頂尖的手機，更是有效率地改變著人類的科技進步。”
- “有關元宇宙的消息在科學界掀起了一陣熱潮。專家指出，它像先進的區塊鏈一樣在科技公司塑造，並且尖端地引領著元宇宙產業的未來。
許多專家正密切關注元宇宙的最新發展。有時候，它不僅是革命性的雲端技術，更是前所未有地實現著人類的科技進步。”

Note:

- Suggest using “string templates” to construct the article paragraphs.
- The font file WCL-07.ttf is placed in the same directory as this HTML file.
- You may modify the HTML and CSS in Test-4-partial.html as needed.

5. (25%) Please develop a basic drawing app (Test-5-partial.html):

- Create the drawing canvas to a 100×100 table. Each cell should be a small square (about 5×5 pixels), and the whole table width should be around 500 pixels.
- Make sure that the user draws only when dragging the mouse (是拖曳，不是移動而已):
 - Press and hold the mouse button down on a cell to start drawing,
 - Move (drag) the mouse over other cells to continue drawing,
 - Release the mouse button to stop drawing.
- Allow users to draw with different colors while dragging:
 - Hold down Ctrl and drag to draw in blue,
 - Hold down Shift and drag to draw in red,
 - Hold down Alt and drag to draw in green,
 - Drag with no modifier key pressed to draw in gray (default color).

Hold *Ctrl* to draw blue. *Shift* to draw red. *Alt* to draw green. (Drag with no key = gray)



- Add a button "Clear Canvas" to clear all colors on the canvas (reset all cells to the original background).

Hints:

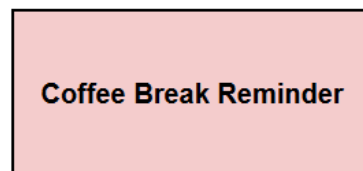
- You can refer to fig. 13.3 in our textbook. (請留意原本範例僅運用 mousemove，但題目要求是拖曳)
- The ctrlKey, shiftKey, and altKey properties of the MouseEvent object should be used.
- You can set an id for each <td> to ease the processing (for example: "canvas-row-col").
- You may use a Boolean flag (e.g., isDragging) to remember whether the mouse button is currently pressed, and combine mousedown, mouseover, and mouseup events to implement the drag-to-draw behavior. (可透過結合 mousedown、mouseover、mouseup 做到接近拖曳的效果)

6. (25%) Please build a small “Virtual Coffee Break Reminder” panel. The panel is used by a busy worker to remember taking short breaks during work. You may refer to Test-6-Partial.html for the basic HTML and CSS structure. You should write your JavaScript code inside the `<script>` section only.

Initial UI:

- A box with the text: “Coffee Break Reminder”.
- Three lines of text under the box:
 - (1) A status message.
 - (2) The total number of times the user has checked the reminder (hovered over the box) (使用者滑過這個 box 幾次).
 - (3) The total number of visits to this page (使用者重載進入過這個頁面幾次).

Virtual Coffee Break Reminder



The reminder is running. Don't forget to take a break!

You have checked this reminder 0 times.

You have opened this reminder 1 times.

Functional Requirements:

- (1) Blinking reminder using `setInterval()`
 - When the page is loaded, use `setInterval()` to change the **background color** of the box every **500 milliseconds** (開始背景顏色持續切換效果).
 - Use this to simulate a “blinking” reminder (cycle through five colors: #ffd966, #ffe599, #f6b26b, #f4cccc, and #c9daf8).
- (2) Pause and resume on mouse events (mouseover/mouseout)
 - When the mouse **moves over** the box (mouseover event):
 - **Stop** the blinking timer (暫停背景顏色持續切換效果).
 - Update the status message to something like: *Enjoy your coffee break now! The reminder is paused.*
 - This hover should be counted in the check count (see Requirement 3).

Virtual Coffee Break Reminder



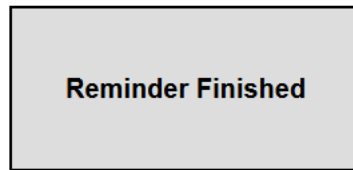
Enjoy your coffee break now! The reminder is paused.

You have checked this reminder 1 times.

You have opened this reminder 1 times.

- When the mouse **moves out** of the box (mouseout event):
 - **Restart** the blinking timer.
 - Update the status message to something like: *The reminder is running. Don't forget to take a break!*
- (3) Store and display check count using *localStorage*
 - Use *localStorage* to store the **total number of times** the user has hovered over the box (across all visits).
 - For each mouseover on the box: read the current value from *localStorage* (use 0 if it does not exist), increase it by 1, save it back to *localStorage*, and display the updated value in the second line, for example: *You have checked this reminder 5 times.*
- (4) Store and display visit count using *localStorage*
 - Use *localStorage* to store the **number of visits** to this Coffee Break Reminder page.
 - When the page is loaded: read the current visit count from *localStorage* (use 0 if it does not exist), increase it by 1, save it back to *localStorage*, display the updated value in the third line, for example: *You have opened this reminder 3 times.*
- (5) Click to “flip” and finish the reminder
 - When the box is **clicked**:
 - Treat this as finishing the reminder (like flipping a card to its back side).
 - **Stop** the blinking timer and **do not restart** it again (停止背景顏色持續切換效果).
 - Change the background color to gray, and change the text to “Reminder Finished”.
 - Update the status message to something like: *The reminder has been finished. Have a nice day!*
 - After the reminder is finished:
 - Further mouseover, mouseout, or click events on the box should not change the color, not restart the timer, and not update the check count.

Virtual Coffee Break Reminder



The reminder has been finished. Have a nice day!

You have checked this reminder 5 times.

You have opened this reminder 2 times.

7. (5%) Please provide some lessons learned from this course and suggestions for this course.